

DATABRICKS DATA + AI PLATFORM

The Brightwave Campaign Desk

Turning recoverable ad spend into revenue —
in minutes, on one governed platform.

RECOVERABLE AD SPEND

\$16.0M

89 Underperforming Campaigns

PROJECTED NET VALUE

\$28.5M

+2.23 ROAS Lift Projected

The mandate: recover the \$16.0M fast — while every AI action stays inside a hard, governed spend cap.

We see the waste. Acting on it takes days.

RECOVERABLE WASTE

\$16.0M

Sitting in 89 underperforming campaigns (avg ~\$180K each).

THE PERFORMANCE GAP

3.7× Gap

Winners return \$4.17 per \$1 spent; underperformers return just \$1.14.

OPERATIONAL COMPLEXITY

Siloed Tools & Teams

Acting means hopping across BI dashboards, spreadsheets, a creative library, and an engineering ticket.

THE COST OF DELAY

Bleeding Budgets

"By the time we move the budget, half the quarter is gone."

The outcome: recovered revenue in minutes

RECOVERABLE REVENUE

\$16.0M

Model projects \$28.5M in net value if every open underperformer is acted on.

SINGLE DECISION IMPACT

+\$416K

Campaign CMP-0000214 achieves a projected +2.23 ROAS lift.

OPERATIONAL SPEED

Minutes

Time-to-action drops dramatically from days to minutes on a single screen.

TRUST & AUDIT

Human-Approved

Every action remains fully audited and requires direct marketer approval.

The Campaign Desk: Visualize → Assist → Act

01 / VISUALIZE

Ranked Performance

Every campaign ranked by recoverable spend, with underperformers and their matching winners, live.



02 / ASSIST

Intelligent Analysis

Ask "why is this losing?" and "what if?"; the agent explains, ranks actions, and drafts an on-brand brief.



03 / ACT

Seamless Execution

Approve, and the decision writes back to the operational store; the queue updates and the KPI ticks down.

One screen. **One agent.** Human in the loop.

Live demo: one question → one approved action

01 / TARGET

Identify Waste

CMP-0000214

Gen Z apparel campaign with **ROAS 1.14** and **\$187K** recoverable spend.

02 / ASSIST

Match Winner

The agent finds high-performing twin **CMP-0000634**.

Boasts **ROAS ~4.96** utilizing social + testimonial creative.

03 / RANK

Evaluate Plays

Ranks opportunities:

- **Replicate: +\$416K**
- Reallocate: +\$171K
- Pause: +\$0.5K

Drafts the optimized brief.

04 / ACT

Instant Execution

Marketer approves. Decision immediately writes back to system.

Queue flips and open underperformers drop **89 → 88**.

Automated workflows with **Human-in-the-Loop** control.

Business value

01 / RETURN

Revenue

Recover **\$16.0M** of wasted spend. **\$28.5M** projected net value across 89 campaigns — a direct lift to the quarterly ROAS the CMO reports to the board.

02 / VELOCITY

Speed

Insight-to-action from **days to minutes**. No tool-hopping, no engineering tickets.

03 / GOVERNANCE

Control

Every action is **human-approved**, backed by a full decision audit trail.

04 / PROTECTION

Cost & Risk

AI spend is **capped and governed** — preventing runaway bills and data leaks.

ROI on the governed AI: **~\$320K net value recovered per campaign** vs cents-to-dollars of capped, metered AI per decision — the AI is a rounding error on the return.

Why Databricks: Context, Control, Cost, Choice

01 / CONTEXT

Context

Connect AI to Brightwave's own campaign and creative data, safely and fast.

02 / CONTROL

Control

Govern what the agent can read, do, and spend; every action audited.

03 / COST

Cost

One control plane for AI spend: budgets, rate limits, full visibility.

04 / CHOICE

Choice

No second database, no reverse-ETL to build or run — Lakebase is the operational store, synced from the lakehouse. Any model, any cloud.

One governed platform. **One control plane for AI cost.**

How it works — **one platform** , end to end

01 / DATA

Lakehouse

Gold campaign & creative data (**2,000** campaigns scored, **400** creatives).

02 / PIPELINE

Lakeflow Sync

Synced to Lakebase — serverless Postgres + Search (BM25 + vector). No separate ops DB, no reverse-ETL.

03 / AGENT

Databricks App

Campaign Desk agent with **5 governed tools**: BI/Genie, Lakebase read, ML ranking, creative search, write-back.

04 / SECURITY

AI Gateway

Every model call routed, guarded, metered; all under **Unity Catalog** governance.

Governed end to end — Unity Catalog · Unity AI Gateway: one control plane for data + AI cost

The 4 C's — delivered by what we built

01 / CONTEXT

Context

Lakebase + Lakebase Search ground the agent in Brightwave's real creatives and campaigns; instant branching powers what-if.

02 / CONTROL

Control

Unity Catalog governs the data; an AI-Gateway guardrail blocks a runaway "dump all data" read; every action is human-approved and audited.

03 / COST

Cost

an AI-Gateway budget caps spend — we enforced a live block at the cap; every call is logged to an inference table.

04 / CHOICE

Choice

the gateway routes the app and the coding agent to different models — swap models without touching the app.

Unity AI Gateway — one control plane for AI cost

01 / GUARDRAIL

Guardrail

An LLM-judge policy blocks all-Lakebase-data reads before they run. **Proven live.**

02 / BUDGET

Budget

A block-usage \$ cap halts spend across the whole workspace. **Proven live** (HTTP 403 at the cap).

03 / TRACE

Trace

Every model call is logged to a governed inference table: app, coding agent, and Slack MCP alike.

04 / COST + CHOICE

Choice + Cost

Route each workload to the right model, and see and cap the bill in one place.

It's real — we built and validated all three layers

01 / Storage & Search

Lakebase

Operational schema (keyed, related tables), hybrid Search (BM25 + vector), copy-on-write branching, scale-to-zero.

02 / Interface

App

The Visualize → Assist → Act desk, retrieving from Lakebase Search and writing approved actions back.

03 / Control Plane

Unity AI Gateway

Governed model service with guardrail, budget, and inference logging — extended to the coding agent + MCP.

✓ **LIVE VERIFICATION:** Every layer executed and verified end-to-end on the live workspace.

Built for the two decisions in this room

CMO / THE BUSINESS

Value & Velocity

- **Recovers real revenue**
\$16.0M recoverable, \$28.5M projected net value.
- **Acts in minutes, not days**
Instantly identify and address campaign performance.
- **Stays in control**
Fully human-approved and auditable at every step.

CDO & CTO / THE PLATFORM

Governance & Openness

- **One governed platform**
Consolidates data, AI, and apps with no operational sprawl.
- **One AI control plane**
Set a hard \$ cap per team — overspend blocked at the cap (proven live), every call metered, so you track cost-per-decision.
- **Any model, any cloud**
No separate operational DB or reverse-ETL to build, secure, and run — one system to secure and operate, not three.

Brightwave + Databricks

**Every ad dollar —
governed,
explained,
and acted on in
minutes.**

01 / THE SOLUTION

Campaign Desk

The Brightwave Campaign Desk, built and deployed on the Databricks Data + AI Platform.